MORPHINE

High-alert drug
Controlled drug

CHEMICAL NAME (4R,4aR,7S,7aR,12bS)-3-methyl-2,3,4,4a,7,7a-hexahydro-1H-4,12-methanobenzofuro[3,2-e]isoquinoline-7,9-diol (as sulfate)

CLASS Strong opioid agonist (phenanthrene alkaloid)

TRADE NAMES MST Continus · Oramorph · Sevredol · Morphine sulfate injection

THERAPEUTIC USES / INDICATIONS

- Severe acute pain: post-operative, trauma, myocardial infarction.
- Chronic cancer pain and palliative care.
- Acute pulmonary oedema (reduces anxiety and preload) — selected cases.
- Breathlessness in palliative care.

PHARMACODYNAMICS

- Agonist at **μ (mu) opioid receptors** in the brain and spinal cord.
- Effects: analgesia, euphoria, sedation, respiratory depression, miosis, cough suppression, ↓ GI motility.
- Causes histamine release → itching, hypotension.

ADMINISTRATION

- **IV:** dilute and give slowly over 4–5 min, titrating to effect.
- **SC / IM, PO** (immediate- or modified-release), PCA pump, epidural/intrathecal.
- Do not crush modified-release tablets.
- Controlled-drug rules: two-nurse check, register entry, witnessed waste.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, vomiting, constipation.
- Drowsiness, dizziness, confusion.
- Itching, sweating, urinary retention, dry mouth.

ADVERSE EFFECTS (SERIOUS)

- Respiratory depression (RR <10/min); apnoea.
- Hypotension, bradycardia.
- Dependence, tolerance, withdrawal.
- Overdose triad: coma, pinpoint pupils, respiratory depression → antidote naloxone.

DOSAGES

- **IV acute pain (adult):** 2.5–5 mg (0.1 mg/kg), titrated every 5–10 min.
- **SC/IM:** 5–10 mg every 4 h.
- **PO immediate-release:** 5–10 mg every 4 h; adjust to response.
- **Children 1–12 yr:** IV 100–200 mcg/kg over ≥5 min, every 4 h if needed (not recommended <1 yr).
- Lower doses in elderly, renal or hepatic impairment.

CONTRAINDICATIONS & CAUTIONS

- Acute respiratory depression; acute severe asthma.
- Head injury or raised intracranial pressure (masks signs).
- Paralytic ileus; risk of ileus.
- **Caution:** COPD, elderly, renal/hepatic impairment, hypothyroidism, pregnancy near delivery.

KEY INTERACTIONS

- **Benzodiazepines, alcohol, sedatives, gabapentinoids:** additive CNS and respiratory depression.
- **MAOIs:** severe reactions; avoid.

PHARMACOKINETICS

- **Onset:** IV 5 min; SC/IM 15–30 min; PO 30–60 min. **Peak:** IV 20 min. **Duration:** 3–5 h (MR 12 h).
- Oral bioavailability ~30% (first-pass). Metabolised in the liver to morphine-6-glucuronide (active).
- Excreted in urine; accumulates in renal failure. **t½:** 2–4 h.

NURSING MANAGEMENT

- Assess pain, RR, sedation score, SpO_2 and BP before and after each dose.
- Withhold and report if RR <10/min or patient is hard to rouse; keep naloxone available.
- Give laxative regularly; monitor bowel movements and urine output.
- Fall precautions; bed rails as per policy.
- Follow controlled-drug procedures.

PATIENT EDUCATION

- Do not drive or use machines if drowsy; avoid alcohol and sleeping pills.
- Take laxatives as prescribed; increase fluids and fibre.
- Do not stop long-term use suddenly.
- Store safely, out of children's reach.

KEY POINTS

- μ-agonist; gold-standard strong opioid.
- Constipation is universal; give laxatives.
- Watch RR and sedation — antidote naloxone.
- Accumulates in renal failure.

SOURCES CHECKED [30] Morphine Sulfate 10mg in 1ml Solution for Injection (UK SmPC) [31] Morphine 10mg/5ml Oral Solution (UK SmPC)
Full references p. 432–457